Morphology as a Lever for Safe Reinforcement Learning: Constrained Co-Design of Body and Controller

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ABSTRACT

Final version: one paragraph, at most 200 words. Content, in order:

- **Problem.** Safe reinforcement learning (RL) optimises the controller for a fixed body, but whether a fast and safe behaviour exists, and whether it can be learned, depends on the body.
- **Approach.** Constrained co-design: an evolved Gaussian-mixture search distribution over ant limb and torso dimensions, a morphology-conditioned PPO-Lagrangian controller trained across a population of bodies simulated side by side on one GPU, one Lagrange multiplier per mixture component, and feasibility-first ranking of designs.
- **Result 1.** On a hazard-field crossing task the co-designed ant is about 55% faster than the nominal ant at a comparable safety cost (1.89 vs 1.20–1.22 m/s). [CHECK] final numbers.
- **Result 2.** Long legs and a small torso evolve consistently; shin length tracks the hazard statistics. Ranking designs by speed alone selects bodies 5× over the budget when the constraint is hard to meet, so the outer loop needs constraint handling too.
- **Result 3.** An open GPU stack for safe RL and safe co-design on MuJoCo Warp: 118–133k steps/s on one RTX 3090, and a 2-billion-step co-design run in about 4 h.
- **Takeaway.** Morphology is a lever for safety, not only for performance.

CCS CONCEPTS

• **Computing methodologies** → **Reinforcement learning**; *Physical simulation*; • **Computer systems organization** → **Evolutionary robotics**.

KEYWORDS

morphology-control co-design, safe reinforcement learning, constrained Markov decision processes, constraint handling, evolutionary robotics, legged locomotion, GPU simulation

DRAFT STATUS AND PLAN (DELETE BEFORE SUBMISSION)

- **What this is.** An outline for the end-of-October report and a later GECCO 2027 submission (Complex Systems track). Tags: [HAVE] result in hand, [RUN] experiment to run, [CHECK] number or claim to verify, [WRITE] prose or figure to produce, [OPTIONAL] stretch.
- **Claim we can make today.** The co-designed ant is about 55% faster than the nominal ant at a comparable episode cost (one seed). **Claim we want.** Co-design moves the safety–performance frontier (faster at equal cost *and* cheaper at equal speed), and the safety constraint changes which body evolves.
- **Every [HAVE] number is preliminary.** They come from the pre-Warp stack (git tag `pre-warp`: MJX-JAX, MuJoCo 3.3.2, solver capped at 4 iterations / 8 line-search steps) with one seed. The same policy runs 4.5% slower on the current stack, so regenerate every table on MuJoCo Warp with the full solver and never mix the two stacks in one table.
- **Experiment queue.** Costs in RTX 3090 GPU-hours on Warp: about 1 h per 300M-step single-body run and 5 h per 2B-step co-design run.
  (1) *P1, needed for any claim (report):* nominal ant with PPO-Lagrangian at $d = 2$, 3 seeds (3 h); co-design at $d = 2$ on the lattice, 3 seeds (15 h); the evolved body retrained from scratch at $d = 2$, 3 seeds (3 h); unconstrained co-design, 3 seeds (15 h), then its body trained with the constraint (3 h).
  (2) *P2, method contribution:* implement feasibility-first ranking; on random fields, speed-only vs feasibility-first, 3 seeds each (30 h); shared vs per-component λ (15 h).
  (3) *P3, headline figure:* budget sweep $d \in \{0, 5, 25\}$ for the nominal ant and for co-design (20–45 h).
  (4) *Optional:* the lasers + mines task (body pressures that oppose each other); same-task throughput against the original mjx-safety-gym, Safety-Gymnasium (CPU) and CRAX.

  About 100–125 GPU-hours in all; roughly a week of wall clock with 4–6 cluster jobs in parallel.
- **Open decisions.**
  (1) Keep actuator rescaling (search shape at constant torque margin), or fix the gear or add an energy cost so that size is not free.
  (2) Widen the design box: 4 of 7 genes sit at the 1.4 edge, and Schaff et al. used 0.5–1.5.
  (3) Envs per GPU: stay at 1024 (as all current runs) or move to 4096. 4096 is 1.75× faster, but co-design iterations must lengthen and learning at that size is unverified.
  (4) Scope of the simulator contribution (Sec. 6.5).

1 INTRODUCTION

Target length: 1 page including Fig. 1

- **Hook.** Legged robots increasingly have to move quickly through places where some contacts are unacceptable: crop

rows they must not trample, clutter they must not disturb, hazards they must not step on. Safe RL formalises this as a constrained Markov decision process (CMDP): maximise return subject to a bound on expected cost [1, 2, 32].

- **The overlooked variable.** Safe RL optimises the *controller* for a *fixed* body. Yet the safety–performance trade-off a robot can reach belongs to body and controller together: morphology shapes which behaviours are possible, robust and easy to learn [4, 31]. Example from our task: legs long enough to clear a hazard turn "weave around it" into "step over it".
- **Research question.** Can optimising morphology improve the safety–performance trade-off beyond what constrained policy optimisation reaches on the nominal body? That is, is morphology a lever for safety and not only for performance?
- **Gap.**
  - Co-design work optimises reward only [16, 17, 36, 38, 44, 49].
  - Where safety appears in co-design, it is a penalty inside the reward [52], the task itself (fall recovery [50]), or a design specification such as velocity, power or payload [23].
  - Safe-RL work keeps the body fixed [7, 21, 24, 32].
  - A systematic citation search (Sec. 2) found no work that trains the controller with constrained RL inside a design loop. [CHECK] rerun before submission.
- **Why it is not just "add a cost term".** There are two coupled constraint problems:
  (1) *Inner.* The budget must hold for every body, not on average over the population the shared policy trains on; otherwise a reckless body can hide behind safe ones.
  (2) *Outer.* Ranking designs by speed assumes the inner loop has already met the budget. When it has not, the search selects fast, unsafe bodies, which we observe (Sec. 6.3). This is the classic constraint-handling problem of evolutionary computation [9, 10, 34], with one twist: feasibility is a noisy estimate that depends on a controller that is still learning.
- **Compute.** Population-based co-design needs billions of simulation steps. GPU simulation makes that practical [11, 29, 51]; we give every body its own lanes in one GPU batch.
- **Contributions.**
  (1) *A safe co-design method.* A constrained bilevel formulation; a Schaff-style Gaussian-mixture design search [36] with a morphology-conditioned PPO-Lagrangian controller; one Lagrange multiplier per mixture component; and feasibility-first design ranking in the outer loop.
  (2) *Evidence that morphology is a safety lever.* On a hazard-field locomotion task the co-designed ant crosses about 55% faster than the nominal ant at comparable constraint cost, and the evolved body changes with the hazard statistics. [HAVE] (1 seed) [RUN] seeds and controls.

Teaser: path-traced frames of the nominal ant (top) and the co-designed ant (bottom) crossing the same hazard field at equal times.

[HAVE] renders: `renders/comparison/evolved_vs_nominal.mp4` and stills; crossing time 9.3 s (nominal) vs 5.4 s (evolved) in these episodes.

Figure 1: Same task, same safety budget, different body. [WRITE] caption with the main numbers once final.

  (3) *A failure mode of naive safe co-design.* With speed-only design ranking, the search returns a body 5× over budget when the constraint is hard to meet (random hazard fields). [HAVE] [RUN] the fix.
  (4) *An open-source GPU stack* for safe RL and safe co-design on MuJoCo Warp, with a different body in every environment lane. It runs 3.1–3.3× faster than the MJX-JAX implementation it extends on the same task (about 19× at equal solver accuracy), with physics equivalence checked; a 2B-step co-design run takes about 4 h on one RTX 3090. [HAVE]

2 RELATED WORK

Target length: 0.75 page

- **Safe RL and its benchmarks.**
  - CMDPs [2]; Lagrangian and primal-dual methods, including PPO-Lagrangian [32].
  - PID Lagrangian [40]: relevant because we observed integral windup of the multiplier (Appendix B).
  - CPO [1], CRPO [48], Sauté RL [39]; surveys [5, 12].
  - Benchmarks: Safety Gym [32], Safety-Gymnasium [21], OmniSafe [22].
  - On accelerators: mjx-safety-gym [43], the MJX port our code extends, and CRAX [42]: 8 tasks at 3 difficulty levels (Point, Ant, Humanoid and other agents), 100–200× faster than CPU benchmarks. Its physics steps on MJX-JAX; it already ships MuJoCo Warp for its pixel renderer (checked October 2026).
  - All of these fix the body.
- **Constrained legged locomotion.** Constraints instead of reward shaping [24], constraints as terminations [7], constrained visual parkour [6], collision-free agile locomotion [19]. All on fixed bodies.
- **Morphology–control co-design.**
  - Evolutionary roots: virtual creatures [38], physical robots [27], soft robots [3, 8].
  - RL-based: learned design distributions [17, 35, 36], graph and transform policies [44, 49], Darwinian evolution with learning [16], co-adaptation [28], universal controllers [15, 20, 25].

  - Constraints so far act on the *design* (fixed parts and allowed ranges [45]; performance specifications [23]) or enter as reward penalties [52]; fall recovery as the objective [50].
  - Literature search (2026-09-29, Semantic Scholar, arXiv and OpenAlex): of 911 papers citing 11 co-design seed papers, 57 mention safety and none trains the controller with constrained RL; of 2,552 papers citing 5 safe-RL seed papers, 21 mention design or morphology and none optimises a body. [CHECK] update.
  - Positioning: the first to put a budgeted, constrained-RL inner loop *and* constraint handling into the design loop.
- **Constraint handling in evolutionary computation.**
  - Penalties, feasibility rules [10], stochastic ranking [34], ε-constrained methods [41]; survey [9].
  - Our outer loop is a search-distribution method in the EDA / natural-evolution-strategy family [26, 46], updated by score-function gradients [47]. Feasibility-first ranking is Deb's rule applied to its utilities.
  - What is new: constraint values are noisy, non-stationary estimates, because the controller co-adapts with the design.
- **GPU simulation.**
  - Brax [11], Isaac Gym [29], MJX and MuJoCo Playground [51], MuJoCo Warp [14], Genesis [13]; massively parallel locomotion RL [33].
  - GPU simulators batch one kinematic structure, which is why our design space is parametric rather than topological (Sec. 7).

3 PROBLEM FORMULATION

Target length: 0.5 page

- **Design space.**
  - $\xi \in \Xi = [0.6, 1.4]^7$: scale factors on the Safety Gym ant [32], namely hip, thigh and shin lengths; hip, thigh and shin radii; and torso radius.
  - Shared by all four legs. Nominal $\xi = \mathbf{1}$ (42.3 kg).
  - The kinematic structure is the same for every ξ (a parametric design space).
- **Actuation.**
  - Actuator gear is rescaled with mass × reach so the torque margin stays constant: we search over shape at constant relative strength.
  - Without this, gear-to-gravity ratios spanned 0.94–3.82 and some bodies could not hold their own legs up.
  - Bodies outside 15–90 kg are rejected.
  - [CHECK] keep or change this (Limitations).
- **Fixed-design CMDP.** For a given ξ, $\max_\pi J_R(\pi;\xi)$ s.t. $J_C(\pi;\xi) \le d$, where J_C is the expected *total* cost of an episode and d the budget.
- **Safe co-design** (bilevel):

$$\max_{\xi\in\Xi} v(\pi^*_\xi;\xi) \quad \text{s.t.} \quad J_C(\pi^*_\xi;\xi) \le d, \qquad \pi^*_\xi \in \arg\max_{\pi:\, J_C(\pi;\xi)\le d} J_R(\pi;\xi), \tag{1}$$

  with v the crossing speed.

- **Amortisation.** One policy $\pi_\theta(a \mid s, \xi)$, conditioned on the design [36], trained on bodies drawn from a search distribution $p_\phi(\xi)$, solves the inner problem for the whole population at once.
- **Why speed and not return.**
  - Return telescopes to distance travelled, so it saturates as soon as bodies arrive. In an early run all 8 bodies scored 22.49–22.50 of a 22.50 ceiling while their arrival times differed by 1.85×.
  - Design score v = net forward displacement per decision. A non-arrival is charged the full 625-decision horizon, so "dash two metres and fall" never scores like a crossing.

4 METHOD

Target length: 1.25 pages including Fig. 2 and Alg. 1

4.1 Population training on one GPU

- Each design iteration samples $K = 16$ designs from p_ϕ and spreads them over 1024 parallel environments in blocks of 64 lanes per design.
- Per-lane MuJoCo models differ only in 14 batched fields (geometry, inertia, gear). The structure is identical, so one compiled step function serves every body: no recompilation between iterations. Batched and individually compiled physics agree to about 10^{-8}.
- The design genes are appended to the observation (243 + 7 = 250 inputs).

4.2 Inner loop: per-component PPO-Lagrangian

- PPO [37] with a separate cost critic; Lagrangian relaxation as in PPO-Lagrangian [32]:

$$\min_{\lambda\ge 0}\max_{\theta}\ \mathbb{E}_{k,\ \xi\sim p^k_\phi}\Big[J_R(\pi_\theta;\xi) - \lambda_k\big(J_C(\pi_\theta;\xi) - d\big)\Big]. \tag{2}$$

- **One multiplier per mixture component** k (contribution). Each sample's cost advantage is weighted by $\lambda_{k(\xi)}$, so the budget has to hold for each subpopulation, not only on average.
- Update: $\lambda_k \leftarrow \mathrm{clip}\left(\lambda_k + \eta_\lambda(\hat{J}^k_C - d),\ 0,\ \lambda_{\max}\right)$ once per mini-batch, with $\eta_\lambda = 1.5 \times 10^{-5}$, $\lambda_{\max} = 3$ (anti-windup) and $\lambda_0 = 0.01$.
- Why the cap: without it, the integral-only multiplier ratcheted up and collapsed a working gait in earlier runs.
- The budget is normalised by the episode length actually observed, so d is an episode total and a policy cannot gain budget by loitering.
- Flips and corridor exits end the episode with a one-off cost of 50. This closes the "end the episode to escape future cost" loophole that we observed with CRPO.

4.3 Outer loop: an evolving design distribution

- **Search distribution (following Schaff et al. [36]).**
  - A mixture of 8 diagonal Gaussians on $[-1, 1]^7$: means drawn from $U(-0.8, 0.8)$, standard deviation 0.577.

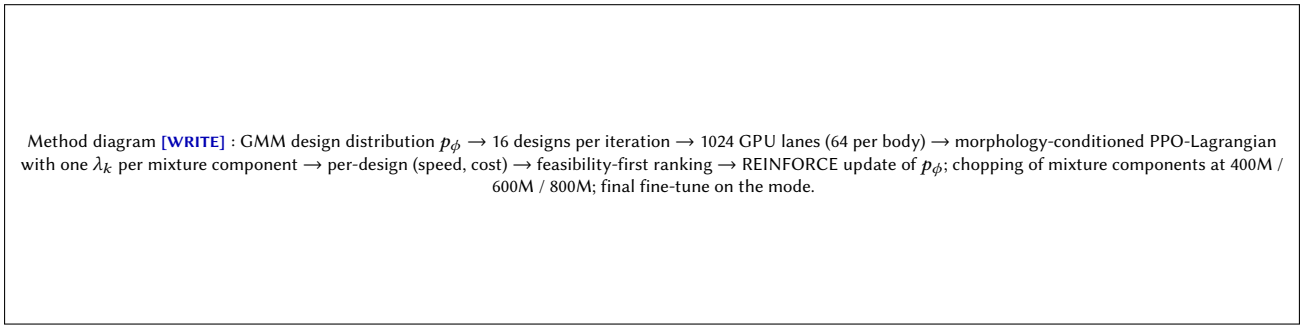


Figure 2: Constrained co-design. Inner loop: a constrained controller for a population of bodies. Outer loop: an evolving design distribution that ranks designs feasibility-first.

  - Mixture weights frozen and uniform.
- **Update.**
  - Score-function gradient through the sampled component only, evaluated at the unclipped sample $\tilde{\xi}$:

$$\nabla_\phi \approx \frac{1}{K} \sum_{i=1}^{K} u_i \nabla_\phi \log p_\phi^{k_i}(\tilde{\xi}_i), \tag{3}$$

    where u_i are scores standardised across the iteration's designs (current) or rank utilities (proposed, Sec. 4.4).
  - Adam with learning rate 0.03.
  - Designs with no completed episode are masked out.
- **Commitment by chopping.** The worst half of the live components is retired at 400M, 600M and 800M steps (8⟶4⟶2⟶1), after a 200M warm-up with the distribution frozen. The last 300M steps fine-tune on the mode.
- **Read as an evolutionary algorithm:** an EDA / NES-like search distribution [26, 46] with truncation selection between subpopulations.
- **Sizing rule.** Each design iteration must contain at least one episode per lane: we use 3.85M steps, about 1.5 episodes. With shorter iterations only flips end episodes early, so scoring is biased against stable bodies.

4.4 Constraint-aware design ranking (proposed)

- [RUN] implement and evaluate.
- Score each design by $(\hat{v}_i, \hat{c}_i)$: the mean speed and mean episode cost of its lanes in the iteration (about 96 episodes per design).
- **Feasibility first** (Deb's rules [10]). Design i ranks above j if:
  - $\hat{c}_i \le d < \hat{c}_j$; or
  - both are feasible and $\hat{v}_i > \hat{v}_j$; or
  - both are infeasible and $\hat{c}_i < \hat{c}_j$.
- Replace the standardised speeds with centred rank utilities (fitness shaping as in NES [46]). Use the same ordering when chopping.
- **Guards.**
  (1) Minimum progress: a body that does not arrive cannot be "feasible", so immobility is never rewarded.
  (2) Noise: a confidence bound $\hat{c}_i + z\,\mathrm{SE}_i \le d$, or stochastic ranking [34].

Algorithm 1: one design iteration [WRITE] typeset properly (e.g. algorithm2e)

(1) Sample K designs $\tilde{\xi}_i \sim p_\phi$ (component k_i); clip into Ξ; build per-lane models.
(2) Reset all lanes (redrawing random hazard fields if enabled).
(3) Run PPO-Lagrangian epochs on the population. Update θ, the critics and λ_k.
(4) From completed episodes, estimate $\hat{v}_i$ and $\hat{c}_i$ for each design.
(5) Rank the designs (speed only, or feasibility first) and compute utilities u_i.
(6) Take a REINFORCE step on ϕ. If a chop is due, retire the worst half of the components and reset the design optimiser.

- **Ablation set:** speed only (Schaff, current); static penalty $v - \mu \max(0, c - d)$; feasibility first. [OPTIONAL] a learned per-design multiplier $\lambda(\xi)$.

4.5 Implementation

- JAX with brax-style PPO; MuJoCo Warp [14] through the MJX interface.
- Everything runs on the device except the design update, which runs on the host between epochs.
- Safe-RL algorithm implementations (PPO-Lagrangian, CRPO, Sauté) adapted from the ss2r and safe-learning codebases. [CHECK] citations.
- Details and measurements in Sec. 6.5 and Appendix D.

5 EXPERIMENTAL SETUP

Target length: 1 page including Tables 1 and 2

5.1 Task: Minefield

- **Corridor.** 11 m long (start $x = -5$, finish line at $x = +6$) and 2 m wide. Leaving it ends the episode with an exit cost of 50.
- **Hazards.**
  - 20 discs of radius 0.16 m in the 10 m × 2 m band.
  - Staggered lattice: 10 columns at 1 m pitch, two discs per column, alternate columns shifted by ±0.21 m.
  - Fixed spawn at the corridor start, so on the lattice every episode has the same layout.

  - Hazards do not collide: they are cost regions, as in Safety Gym.
- **Variants.**
  - Radius 0.23 m: 2.07× the area, covering 16.6% of the band instead of 8.0%.
  - Random placement: rejection-sampled at every reset, so a 2B-step run sees about 530k distinct fields. Here safety cannot be a memorised stride.
- **The constraint is feasible.** A kinematic check shows a zero-cost crossing exists for the *nominal* ant on the lattice: 77% of torso positions in the band admit a cost-free four-foot stance, and they connect start to finish. Residual cost therefore reflects what the policy learned, not infeasibility.

5.2 Robot and observations

- **Robot.** The Safety Gym ant: 8 actuators, 42.3 kg, condim 3.
- **Timing.** RK4 with $\Delta t = 0.01$ s and 2 substeps; action repeat 4, so one decision every 0.08 s. An episode is at most 625 decisions (50 s).
- **Observation (243 values).**
  - Proprioception and goal sensing (31).
  - A 16-bin hazard lidar.
  - A per-foot 7×7 grid giving the cost that foot would incur if placed at each cell (4×49).
  - [CHECK] the breakdown of the 31.
- **Co-design input.** The 7 design genes are added for co-design.
- **Why the foot grid is there:** without it the Lagrangian could not reduce cost by improving foot placement (Appendix B).

5.3 Reward, cost and budget

- **Reward.** Forward progress Δx, plus goal-distance shaping, plus an upright bonus of 0.0002 per step. The episode ends on arrival, flip, exit or after 625 decisions.
- **Cost per step.**
  - For each hazard: linear penetration $\text{clip}(1 - \delta/R, 0, 1)$ of the closest *grounded* body part, where δ is the surface distance to the hazard centre and "grounded" means the part's lowest point is within 2 cm of the floor. Summed over hazards.
  - Plus a one-off 50 for a flip or exit.
  - Graded like the hazard costs of Safety Gym and CRAX [32, 42].
- **Budget.** d is the expected episode total. The main experiments use $d = 2$, a strict budget.
- [CHECK] report what a careless crossing costs on the 2 cm scale, to put $d = 2$ in context.

5.4 Training and evaluation

- **Training.** Hyperparameters in Table 2. Co-design runs take 2B steps: 520 design iterations of 3.85M steps. Single-body runs take 250–300M steps.
- **Evaluation.**
  - 512 stochastic episodes (128 environments × 4) on held-out layouts.
  - Metrics: return, speed (m/s), cost (per-environment mean ± SE, $n = 128$), hazard steps (steps with any grounded part inside a disc), flipped %, out-of-bounds %, and the share of environments under budget.
- **Pitfall to state.** The trainer's periodic evaluations use 16 fixed random bodies, not the evolved one. Final bodies are evaluated separately with `scripts/eval_design.py`.
- **Seeds.** 3 per co-design condition and 5 per single-body condition. [RUN]
- **Hardware.** RTX 3090 (cluster) and an RTX 4050 laptop. A 2B-step co-design run takes about 4–5 h on Warp, against 13.9 h on MJX-JAX.

6 RESULTS

Target length: 2.5 pages

6.1 Morphology improves the safety–performance trade-off

- **Headline (Table 3).** At budget 2 the evolved body crosses at 1.89 m/s with cost 2.55 ± 0.34, and 66% of environments stay under budget. The nominal ant reaches 1.20 m/s at cost 1.9 ($d = 0$) and 1.22 m/s at cost 2.7 ($d = 5$). That is about 55% faster at comparable total cost. [HAVE] (pre-Warp, 1 seed)
- **Teaser numbers.** In the rendered episodes the crossing takes 5.4 s against 9.3 s (Fig. 1).
- **State the claim precisely.** Today it is "faster at matched cost", not "safer".
  - The evolved body's *hazard* cost alone is about 1.48, against 0.65 ($d = 0$) and 1.1 ($d = 5$) for the nominal ant.
  - It spends less of its budget on flips and exits (1.07, against 1.25 and 1.6).
  - "Safer" needs the frontier: lower cost at equal speed, from the budget sweep (Fig. 3).
- **Confounds to close.**
  1. Seeds. [RUN]
  2. The nominal ant at exactly $d = 2$. [RUN]
  3. Training budget: the nominal ant had 250M single-body steps; co-design had 2B steps in total, ending with a 300M fine-tune on the evolved body. Condition E (evolved body trained from scratch) separates "better body" from "more training". [RUN]
  4. Evaluation protocol: the nominal numbers are means of the trainer's last three evaluations, the evolved ones come from the 512-episode protocol. Re-evaluate the nominal checkpoints with the same script. [CHECK]

6.2 What evolves

- **Lattice, radius 0.16** (Table 4). Hip, thigh and shin lengths ×1.40 and hip radius ×1.40, all at the box edge; thigh radius ×0.94, shin ×0.83, torso ×0.72. Mass 38.5 kg (nominal 42.3); gear 191 (nominal 150). [HAVE]
- **Consistent across all three regimes:** hip and thigh at the length limit, thick hips, a small torso.

Table 1: Experimental conditions. Hazards: L = lattice, R = random placement; radius in metres. Unless noted, the outer loop ranks designs by speed and the inner loop is PPO-Lagrangian with one λ per mixture component.

ID	Body	Inner loop	Outer ranking	Hazards	Question	Status
N	nominal, fixed	PPO-Lag, $d \in \{0, 2, 5\}$	–	L 0.16	fixed-body baseline	$d = 0, 5$ [HAVE] $d = 2$ [RUN]
CD	co-designed	PPO-Lag, $d = 2$	speed	L 0.16	main method	[HAVE] 1 seed
CD-0.23	co-designed	PPO-Lag, $d = 2$	speed	L 0.23	hazard size	[HAVE] 1 seed
CD-R	co-designed	PPO-Lag, $d = 2$	speed	R 0.16	hard constraint	[HAVE] 1 seed
CD-R-FF	co-designed	PPO-Lag, $d = 2$	feasibility first	R 0.16	outer-loop constraints	[RUN]
E	evolved, fixed	PPO-Lag, $d = 2$ (scratch)	–	L 0.16	body vs. co-training	[RUN]
U	co-designed	none (unconstrained)	speed	L 0.16	does safety shape the body?	[RUN]
U→S	U's body, fixed	PPO-Lag, $d = 2$	–	L 0.16	evolve, then add safety	[RUN]
CD-shared	co-designed	PPO-Lag, one λ	speed	L 0.16	per-component λ	[RUN]
Sweep	both	PPO-Lag, $d \in \{0, 5, 25\}$	speed or FF	L 0.16	frontier (headline)	[RUN]
UNI	uniform designs	PPO-Lag, $d = 2$	none; pick best	L 0.16	is design learning needed?	[OPTIONAL]

Table 2: Hyperparameters (all runs). [CHECK] against the launchers before submission.

Parallel environments	1024 (64 per design, 16 designs)
Policy / value networks	MLP 256 × 4 / 256 × 5
Learning rate, entropy cost, clip	10^{-4}, 10^{-4}, 0.3
Discount (reward / cost)	0.97 / 0.9
Unroll, batch, minibatches	10, 32, 32
Multiplier lr / cap / initial value	1.5×10^{-5} / 3.0 / 0.01
Flip / exit cost	50 / 50
Design components / lr / std init	8 / 0.03 (Adam) / 0.577
Design iteration length	3.85M steps (≈1.5 episodes per lane)
Warm-up / chops / fine-tune	200M / 400, 600, 800M / last 300M
Total steps (co-design / single body)	2B / 250–300M
Matmul precision	float32

Headline figure [RUN] : speed (m/s, y) against episode cost (x, log scale) for the nominal ant across budgets $d \in \{0, 2, 5, 25\}$ (the fixed-body frontier) and for co-designed bodies at the same budgets; budget lines dashed; error bars over seeds. The claim holds if the co-design frontier dominates.

Figure 3: Safety–performance frontier: fixed body vs. co-designed body.

- **What changes:** shin length, 1.40 (lattice 0.16) → 1.22 (lattice 0.23) → 1.03 (random). One seed each, so a hint only.
- **Interpretation** (to test):
  - Long legs give clearance to step over discs, and longer strides.
  - A small torso lowers mass and inertia.
  - On random fields, shorter shins may favour precise placement over clearance.
- **Gait analysis to back it** [RUN] : share of time the torso is above a disc, foot-placement histograms relative to discs, step length.

Gene trajectories [WRITE] : mode of each mixture component across training (7 panels, or 2D projections), with chop events at 400/600/800M and the box limits marked. Data: the design lines in the co-design logs and the `design_state.npz` checkpoints.

Figure 4: How the design distribution moves and commits.

- **Weakness to own:** 4 of 7 genes end at the upper bound, so the bounds bind (Sec. 7).
- **Fig. 4:** the mode of each component across training, with chop events. [HAVE] design logs; [WRITE] plot.

6.3 The outer loop needs constraint handling

- **Random hazard fields** (Table 5). The evolved body crosses at 1.48 m/s but at cost 10.15 ± 0.60, 5× the budget. No environment stays under budget. [HAVE]
- **Training trace.** Every component's λ was pinned at the cap (3.0), with the constraint at −0.71: the inner loop could not meet $d = 2$ for any subpopulation. [HAVE]
- **Mechanism.** Speed-only ranking kept favouring the fastest designs, which were also the most violating. The search had no reason to prefer a feasible body.
- **On the lattice** the same loop happened to settle near the budget (2.55 ± 0.34 and 2.31 ± 0.20; both +1.5–1.6 SE over d, not significant at 2 SE). Feasibility there came from the inner loop, not from the search.
- **The fix:** feasibility-first ranking, CD-R vs CD-R-FF, 3 seeds each (Fig. 5). Report the final body, the cost trace, and whether the search finds a feasible body or proves none is learnable. [RUN]
- **The per-component λ ablation** (CD vs CD-shared) also belongs here: with one shared λ, does a reckless component get carried? [RUN]

Table 3: Evaluation at budget $d = 2$ on the lattice field (radius 0.16). Cost is the per-environment episode total (± SE, $n = 128$); "under budget" is the share of environments whose mean cost is $\leq d$. [HAVE] rows are from the pre-Warp stack with one seed and must be regenerated. [CHECK] the nominal rows are means of the trainer's last three evaluations, not the 512-episode protocol, and the $d = 5$ run had not converged.

Body	Training	Return	Speed (m/s)	Cost	Hazard steps	Flipped (%)	OOB (%)	Under budget (%)	Status
Nominal	PPO-Lag $d = 0$, 250M	11.01	1.20	1.9	3.5	1.4	1.1	–	[HAVE] [CHECK]
Nominal	PPO-Lag $d = 5$, 250M	10.99	1.22	2.7	5.1	2.1	1.1	–	[HAVE] [CHECK]
Nominal	PPO-Lag $d = 2$								[RUN]
Co-designed	CD, 2B	11.09	1.89	2.55 ± 0.34	6.6	0.4	1.8	66	[HAVE]
Co-designed, fixed	E, from scratch								[RUN]
Unconstrained design	U→S								[RUN]

Table 4: Evolved designs (scale factors on the nominal ant). Hip = `aux`, thigh = `leg`, shin = `ankle` segments of the Safety Gym ant. [HAVE] (pre-Warp, 1 seed).

	Length			Radius			Torso	Mass	Gear
Run	hip	thigh	shin	hip	thigh	shin	radius	(kg)	
Nominal	1.00	1.00	1.00	1.00	1.00	1.00	1.00	42.3	150
L 0.16	1.40	1.40	1.40	1.40	0.94	0.83	0.72	38.5	191
L 0.23	1.40	1.40	1.22	1.31	0.76	0.96	0.60	46.2	219
R 0.16	1.40	1.40	1.03	1.32	0.82	0.99	0.80	42.5	193

[RUN] : episode cost of the designs sampled at each iteration (median and IQR) against training steps, speed-only vs feasibility-first on random fields, with the budget line; inset: per-component λ (pinned at the cap under speed-only).

Figure 5: Speed-only design ranking drifts to infeasible bodies; feasibility-first ranking does not.

6.4 Does the safety constraint shape the body?

- This is the core of the "lever" claim. If unconstrained co-design evolves the same body, safety is a by-product of speed, not a design driver.
- **Compare** U against CD (genes and performance), U→S against E (is evolving *under* the constraint better than evolving first and adding safety later?), and the evolved genes against the budget across the sweep. [RUN]
- **Honest expectation.** On this task speed and safety probably pull the same way: long legs help both. The bodies may differ little, which motivates a task with opposing pressures (Sec. 7).

6.5 Simulation stack: throughput and fidelity

- **Throughput** (Table 6). On an RTX 3090 at 1024 environments:
  - 38.2k → 118.3k steps/s for a single body (3.1×) and 40.4k → 133.3k for co-design with 16 bodies (3.3×), against the MJX-JAX backend with its solver capped for speed. [HAVE]
  - At equal solver accuracy (both run to convergence) the laptop shows 3.8k → 73.3k training steps/s (19×), and 4.4k → 140k steps/s for pure stepping (32×). [HAVE] [CHECK] repeat on the 3090
- **Why it is faster.** MJX-JAX pays for every *possible* contact because of static array shapes (about 188 constraint rows per environment here), and its line search runs a fixed number of steps. Warp only processes active contacts (at most 20 rows per environment) and stops once every world has converged [14, 51].
- **Scaling** (Fig. 6): 4096 environments give 220k (single) and 233k (co-design) steps/s, 1.75× the 1024-environment rate. Learning at that size is unverified, and co-design iterations must lengthen to keep at least one episode per lane. [HAVE]
- **Fidelity** (Appendix D).
  - Old and new MuJoCo agree to float32 rounding once the solver converges.
  - Warp and JAX agree to about 10^{-7}, including with a different body in every lane.
  - The vase cost, re-implemented with contact sensors, matches the original step for step.
  - A 300M-step learning check reproduces the old baseline: 1.52 vs 1.45 m/s, cost 26.3 vs 27.4 at $d = 25$. [HAVE]
- **Wall clock.** A 2B-step co-design run drops from 13.9 h to about 4 h of training, which is what makes seeds and sweeps affordable.
- **Positioning** (decision pending).
  - CRAX [42] already provides a broad MJX safe-RL suite, including Ant tasks.
  - The Warp backend alone is a temporary difference: CRAX already depends on MuJoCo Warp, and our own switch took about 300 changed lines in 10 files.
  - Our claim is therefore neither breadth nor the backend. It is (i) a different body in every lane of one batch, with constraint costs computed on the GPU, (ii) the safe co-design loop, and (iii) validated equivalence of the Warp port, including per-lane bodies.
  - [OPTIONAL] same-task throughput: CRAX's Ant goal task against ours; Safety-Gymnasium on CPU.

Table 5: Evolved bodies under three hazard regimes, budget $d = 2$, 512 episodes each. [HAVE] rows: pre-Warp stack, one seed.

Hazards	Outer ranking	Return	Speed (m/s)	Cost	Hazard steps	Flipped (%)	OOB (%)	Under budget (%)	Status
Lattice, $r = 0.16$	speed	11.09	1.89	2.55 ± 0.34	6.6	0.4	1.8	66	[HAVE]
Lattice, $r = 0.23$	speed	11.21	1.63	2.31 ± 0.20	25.3	0.4	0.0	60	[HAVE]
Random, $r = 0.16$	speed	10.88	1.48	10.15 ± 0.60	25.8	1.4	4.9	0	[HAVE]
Random, $r = 0.16$	feasibility first								[RUN]

Table 6: Training throughput (environment steps per second, compilation excluded). Single = one nominal body; co-design = 16 bodies. Solver: *capped* = 4 iterations / 8 line-search steps (the MJX-JAX speed setting), *full* = MuJoCo defaults, run to convergence. [CHECK] re-measure all cells with one script on the 3090.

	RTX 4050 laptop		RTX 3090	
Backend, solver	single	co-design	single	co-design
MJX-JAX, full	3.8k	–	[RUN]	–
MJX-JAX, capped	13.8k	19.0k	38.2k	40.4k
Warp, full	73.3k	100.0k	118.3k	133.3k
Warp, full, 4096 envs	–	–	220k	233k

Laptop single-body runs used 512 environments; all other cells 1024 unless noted.

[HAVE] data, [WRITE] plot: steps/s vs number of environments (1024–8192) on the 3090, Warp, single and co-design; MJX-JAX points for reference. GPU utilisation read 100% at every size, so small batches under-fill the GPU.

Figure 6: Throughput scaling with environments per GPU.

7 DISCUSSION

Target length: 0.75 page

- **Why the body helps.**
  - The nominal ant *can* cross at zero cost (Sec. 5), but constrained RL does not find that precise foot placement.
  - A body with more clearance and a longer stride widens the margin, so the safe crossing becomes easier to learn.
  - Hypothesis: morphology helps safety mainly through *learnability*, not feasibility. Test: rerun the kinematic stance check for the evolved bodies (cheap, CPU only). [RUN]
- **Constraint handling is part of the method, not a detail.** The inner loop protects safety only when it succeeds; the outer loop has to know when it did not.
- **Limitations.**
  (1) One seed per condition and a pre-Warp stack so far.
  (2) Bounds bind: 4 of 7 genes end at the box edge.
  (3) Size is free: actuators are rescaled, so bigger limbs cost no relative strength.
  (4) Speed and safety point the same way on this task, so there is no tension to resolve.
  (5) On the lattice, safety can be a memorised stride; random fields are the hard case, and they fail without feasibility-first ranking.
  (6) Parametric design only. GPU simulators batch one kinematic structure, so topology needs a superset model or per-world kinematic trees (a draft MuJoCo Warp feature).
  (7) Simulation only; no sim-to-real.
  (8) Designs are scored through a shared policy, which generalises by interpolation (novel bodies inside the trained mass range: 7/7 worked) but not by extrapolation (outside it: 3/7).
- **Future work.**
  - A lasers + mines task: beams at random heights. Long legs step over mines but hit beams; low bodies duck beams but must weave between mines, so the best body depends on the mix and on the budget.
  - A payload (tipping or jerk cost) and an energy cost.
  - Topology through a superset ant.
  - A real quadruped model (e.g. from MuJoCo Menagerie).
  - Quality diversity: MAP-Elites over morphology with safety descriptors [30]; CMA-ES as an alternative outer loop [18].

8 CONCLUSION

Target length: 0.25 page

- Safety is a property of the body as well as of the controller.
- Co-designing morphology under a safety constraint found an ant that is about 55% faster at comparable cost. [CHECK] final wording once the frontier is in.
- Safe co-design needs constraint handling at both levels: per-component multipliers inside, feasibility-first ranking outside.
- A GPU stack makes this affordable: billions of steps in hours on one consumer GPU.

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REFERENCES

[1] Joshua Achiam, David Held, Aviv Tamar, and Pieter Abbeel. 2017. Constrained Policy Optimization. In *Proceedings of the 34th International Conference on Machine Learning (ICML) (PMLR, Vol. 70)*. 22–31.
[2] Eitan Altman. 1999. *Constrained Markov Decision Processes.* Chapman and Hall/CRC.

[3] Jagdeep Bhatia, Holly Jackson, Yunsheng Tian, Jie Xu, and Wojciech Matusik. 2021. Evolution Gym: A Large-Scale Benchmark for Evolving Soft Robots. In *Advances in Neural Information Processing Systems (NeurIPS)*.

[4] Josh Bongard. 2011. Morphological Change in Machines Accelerates the Evolution of Robust Behavior. *Proceedings of the National Academy of Sciences* 108, 4 (2011), 1234–1239.

[5] Lukas Brunke, Melissa Greeff, Adam W. Hall, Zhaocong Yuan, Siqi Zhou, Jacopo Panerati, and Angela P. Schoellig. 2022. Safe Learning in Robotics: From Learning-Based Control to Safe Reinforcement Learning. *Annual Review of Control, Robotics, and Autonomous Systems* 5 (2022), 411–444.

[6] Elliot Chane-Sane, Joseph Amigo, Thomas Flayols, Ludovic Righetti, and Nicolas Mansard. 2024. SoloParkour: Constrained Reinforcement Learning for Visual Locomotion from Privileged Experience. In *Conference on Robot Learning (CoRL)*.

[7] Elliot Chane-Sane, Pierre-Alexandre Léziart, Thomas Flayols, Olivier Stasse, Philippe Souères, and Nicolas Mansard. 2024. CaT: Constraints as Terminations for Legged Locomotion Learning. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*.

[8] Nick Cheney, Robert MacCurdy, Jeff Clune, and Hod Lipson. 2013. Unshackling Evolution: Evolving Soft Robots with Multiple Materials and a Powerful Generative Encoding. In *Proceedings of the Genetic and Evolutionary Computation Conference (GECCO)*. 167–174.

[9] Carlos A. Coello Coello. 2002. Theoretical and Numerical Constraint-Handling Techniques Used with Evolutionary Algorithms: A Survey of the State of the Art. *Computer Methods in Applied Mechanics and Engineering* 191, 11–12 (2002), 1245–1287.

[10] Kalyanmoy Deb. 2000. An Efficient Constraint Handling Method for Genetic Algorithms. *Computer Methods in Applied Mechanics and Engineering* 186, 2–4 (2000), 311–338.

[11] C. Daniel Freeman, Erik Frey, Anton Raichuk, Sertan Girgin, Igor Mordatch, and Olivier Bachem. 2021. Brax – A Differentiable Physics Engine for Large Scale Rigid Body Simulation. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*.

[12] Javier García and Fernando Fernández. 2015. A Comprehensive Survey on Safe Reinforcement Learning. *Journal of Machine Learning Research* 16, 42 (2015), 1437–1480.

[13] Genesis Authors. 2024. Genesis: A Generative and Universal Physics Engine for Robotics and Beyond. Software. https://github.com/Genesis-Embodied-AI/Genesis

[14] Google DeepMind and NVIDIA. 2025. MuJoCo Warp: GPU-Optimized MuJoCo for NVIDIA Hardware. Software. https://github.com/google-deepmind/mujoco_warp

[15] Agrim Gupta, Linxi Fan, Surya Ganguli, and Li Fei-Fei. 2022. MetaMorph: Learning Universal Controllers with Transformers. In *International Conference on Learning Representations (ICLR)*.

[16] Agrim Gupta, Silvio Savarese, Surya Ganguli, and Li Fei-Fei. 2021. Embodied Intelligence via Learning and Evolution. *Nature Communications* 12 (2021), 5721.

[17] David Ha. 2019. Reinforcement Learning for Improving Agent Design. *Artificial Life* 25, 4 (2019), 352–365.

[18] Nikolaus Hansen and Andreas Ostermeier. 2001. Completely Derandomized Self-Adaptation in Evolution Strategies. *Evolutionary Computation* 9, 2 (2001), 159–195.

[19] Tairan He, Chong Zhang, Wenli Xiao, Guanqi He, Changliu Liu, and Guanya Shi. 2024. Agile But Safe: Learning Collision-Free High-Speed Legged Locomotion. In *Robotics: Science and Systems (RSS)*.

[20] Wenlong Huang, Igor Mordatch, and Deepak Pathak. 2020. One Policy to Control Them All: Shared Modular Policies for Agent-Agnostic Control. In *Proceedings of the 37th International Conference on Machine Learning (ICML) (PMLR, Vol. 119)*.

[21] Jiaming Ji, Borong Zhang, Jiayi Zhou, Xuehai Pan, Weidong Huang, Ruiyang Sun, Yiran Geng, Yifan Zhong, Juntao Dai, and Yaodong Yang. 2023. Safety-Gymnasium: A Unified Safe Reinforcement Learning Benchmark. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*.

[22] Jiaming Ji, Jiayi Zhou, Borong Zhang, Juntao Dai, Xuehai Pan, Ruiyang Sun, Weidong Huang, Yiran Geng, Mickel Liu, and Yaodong Yang. 2024. OmniSafe: An Infrastructure for Accelerating Safe Reinforcement Learning Research. *Journal of Machine Learning Research* 25, 285 (2024), 1–6.

[23] Amogh Joshi and Kaushik Roy. 2026. GLAMDRING: Gait Learning And Morphology co-Design via Reinforcement LearnING of CPGs. arXiv:2609.19452.

[24] Yunho Kim, Hyunsik Oh, Jeonghyun Lee, Jinhyeok Choi, Gwanghyeon Ji, Moonkyu Jung, Donghoon Youm, and Jemin Hwangbo. 2024. Not Only Rewards but Also Constraints: Applications on Legged Robot Locomotion. *IEEE Transactions on Robotics* 40 (2024), 2984–3003. https://doi.org/10.1109/TRO.2024.3400935

[25] Vitaly Kurin, Maximilian Igl, Tim Rocktäschel, Wendelin Boehmer, and Shimon Whiteson. 2021. My Body is a Cage: the Role of Morphology in Graph-Based Incompatible Control. In *International Conference on Learning Representations (ICLR)*.

[26] Pedro Larrañaga and Jose A. Lozano (Eds.). 2002. *Estimation of Distribution Algorithms: A New Tool for Evolutionary Computation*. Kluwer Academic Publishers.

[27] Hod Lipson and Jordan B. Pollack. 2000. Automatic Design and Manufacture of Robotic Lifeforms. *Nature* 406 (2000), 974–978.

[28] Kevin Sebastian Luck, Heni Ben Amor, and Roberto Calandra. 2019. Data-efficient Co-Adaptation of Morphology and Behaviour with Deep Reinforcement Learning. In *Conference on Robot Learning (CoRL) (PMLR, Vol. 100)*.

[29] Viktor Makoviychuk, Lukasz Wawrzyniak, Yunrong Guo, Michelle Lu, Kier Storey, Miles Macklin, David Hoeller, Nikita Rudin, Arthur Allshire, Ankur Handa, and Gavriel State. 2021. Isaac Gym: High Performance GPU-Based Physics Simulation for Robot Learning. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*.

[30] Jean-Baptiste Mouret and Jeff Clune. 2015. Illuminating Search Spaces by Mapping Elites. arXiv:1504.04909.

[31] Rolf Pfeifer and Josh Bongard. 2006. *How the Body Shapes the Way We Think: A New View of Intelligence*. MIT Press.

[32] Alex Ray, Joshua Achiam, and Dario Amodei. 2019. *Benchmarking Safe Exploration in Deep Reinforcement Learning*. Technical Report. OpenAI. https://cdn.openai.com/safexp-short.pdf

[33] Nikita Rudin, David Hoeller, Philipp Reist, and Marco Hutter. 2022. Learning to Walk in Minutes Using Massively Parallel Deep Reinforcement Learning. In *Conference on Robot Learning (CoRL) (PMLR, Vol. 164)*. 91–100.

[34] Thomas Philip Runarsson and Xin Yao. 2000. Stochastic Ranking for Constrained Evolutionary Optimization. *IEEE Transactions on Evolutionary Computation* 4, 3 (2000), 284–294.

[35] Charles Schaff and Matthew R. Walter. 2022. N-LIMB: Neural Limb Optimization for Efficient Morphological Design. arXiv:2207.11773.

[36] Charles Schaff, David Yunis, Ayan Chakrabarti, and Matthew R. Walter. 2019. Jointly Learning to Construct and Control Agents using Deep Reinforcement Learning. In *IEEE International Conference on Robotics and Automation (ICRA)*. 9798–9805.

[37] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. 2017. Proximal Policy Optimization Algorithms. arXiv:1707.06347.

[38] Karl Sims. 1994. Evolving Virtual Creatures. In *Proceedings of the 21st Annual Conference on Computer Graphics and Interactive Techniques (SIGGRAPH)*. 15–22.

[39] Aivar Sootla, Alexander I. Cowen-Rivers, Taher Jafferjee, Ziyan Wang, David H. Mguni, Jun Wang, and Haitham Bou-Ammar. 2022. Sauté RL: Almost Surely Safe Reinforcement Learning Using State Augmentation. In *Proceedings of the 39th International Conference on Machine Learning (ICML) (PMLR, Vol. 162)*. 20423–20443.

[40] Adam Stooke, Joshua Achiam, and Pieter Abbeel. 2020. Responsive Safety in Reinforcement Learning by PID Lagrangian Methods. In *Proceedings of the 37th International Conference on Machine Learning (ICML) (PMLR, Vol. 119)*. 9133–9143.

[41] Tetsuyuki Takahama and Setsuko Sakai. 2006. Constrained Optimization by the ε Constrained Differential Evolution with Gradient-Based Mutation and Feasible Elites. In *IEEE Congress on Evolutionary Computation (CEC)*. 1–8.

[42] Tristan Tomilin, Mourad Boustani, Mickey Beurskens, and Thiago D. Simão. 2026. CRAX: Fast Safe Reinforcement Learning Benchmarking. arXiv:2606.20376.

[43] Max van der Hart. 2025. mjx-safety-gym: MJX Implementation of OpenAI Safety Gym. Software, PyPI package version 0.1.6. https://github.com/mvdhart/mjx-safety-gym

[44] Tingwu Wang, Yuhao Zhou, Sanja Fidler, and Jimmy Ba. 2019. Neural Graph Evolution: Towards Efficient Automatic Robot Design. In *International Conference on Learning Representations (ICLR)*.

[45] Zhiquan Wang, Bedrich Benes, Ahmed H. Qureshi, and Christos Mousas. 2022. Co-design of Embodied Neural Intelligence via Constrained Evolution. arXiv:2205.10688.

[46] Daan Wierstra, Tom Schaul, Tobias Glasmachers, Yi Sun, Jan Peters, and Jürgen Schmidhuber. 2014. Natural Evolution Strategies. *Journal of Machine Learning Research* 15, 27 (2014), 949–980.

[47] Ronald J. Williams. 1992. Simple Statistical Gradient-Following Algorithms for Connectionist Reinforcement Learning. *Machine Learning* 8 (1992), 229–256.

[48] Tengyu Xu, Yingbin Liang, and Guanghui Lan. 2021. CRPO: A New Approach for Safe Reinforcement Learning with Convergence Guarantee. In *Proceedings of the 38th International Conference on Machine Learning (ICML) (PMLR, Vol. 139)*. 11480–11491.

[49] Ye Yuan, Yuda Song, Zhengyi Luo, Wen Sun, and Kris M. Kitani. 2022. Transform2Act: Learning a Transform-and-Control Policy for Efficient Agent Design. In *International Conference on Learning Representations (ICLR)*.
[50] Bo Yue, Sheng Xu, Kui Jia, and Guiliang Liu. 2025. Toward Humanoid Brain-Body Co-design: Joint Optimization of Control and Morphology for Fall Recovery. arXiv:2510.22336.
[51] Kevin Zakka, Baruch Tabanpour, Qiayuan Liao, Mustafa Haiderbhai, Samuel Holt, Jing Yuan Luo, Arthur Allshire, Erik Frey, Koushil Sreenath, Lueder A. Kahrs, Carmelo Sferrazza, Yuval Tassa, and Pieter Abbeel. 2025. MuJoCo Playground. arXiv:2502.08844.
[52] Yi Zhang, Yue Xie, Tao Sun, and Fumiya Iida. 2025. Co-design is Powerful and Not Free. arXiv:2510.08368.

For the October report these sections are part of the main text. For GECCO they move to supplementary material.

A HYPERPARAMETERS AND REPRODUCIBILITY

- Full flag listing per condition, from the launchers (`cluster/ant_morphology_codesign.sbatch`: knobs BUDGET, HAZARD_SIZE, HAZARD_PLACEMENT, PENALIZER). [WRITE]
- **Code.** Public repository (anonymised for review). Tag `pre-warp` reproduces every [HAVE] number; `main` is the Warp stack.
- **Evaluation.** `scripts/eval_design.py` (evolved bodies), `scripts/eval_checkpoint.py` (single bodies).
- **Checkpoints.** Run names: `ant_codesign_lagrangian_b2_grid7_2000M` and its `_hz0.23` and `_random` variants; `ant_minefield_final_eps2_b{0,5}_250M`.

B DEVELOPMENT HISTORY AND NEGATIVE RESULTS

Report only. The decisions that shaped the final recipe, each backed by a measurement.

- **Goal-reaching was unlearnable for the ant.** With the goal in a random direction, a scripted gait that walks several metres earns +0.078±0.672 return over 16 seeds. Hence the corridor task, with reward linear in forward progress.
- **Policy capacity mattered most early on.** Widening the policy from 32 × 4 to 256 × 4 cut cost 2.7× at the same reward.
- **Constraint mechanisms first "succeeded" by doing less of the task.**
  - CRPO: cost per step *doubled* (0.025 → 0.050) while episode cost fell, because episodes got shorter (2035 → 738 steps).
  - Sauté: the budget was not a lever (tightening it tenfold, 500 → 50, did not lower cost: 84.9 vs 95.7); penalty 5 cut cost 43% and reward 42% together.
  - Lagrangian without a cap: the multiplier wound up and the policy met the budget by walking 3 m and flipping.
- **Fixes, each measured.**
  - A flip/exit cost.
  - A graded penetration cost (Safety Gym / CRAX style).
  - A budget normalised by the observed episode length.
  - A multiplier cap.
  - *The per-foot hazard grid* (50M-step arms, one seed each). Without the grid, the Lagrangian ended *above* its unconstrained control (cost 78.6 vs 63.1). With it, the Lagrangian cut cost below its unconstrained control: by 12% from scratch (44.8 vs 51.1), and by 35% when warm-started from that control (33.0 vs 51.1), with the crossing intact.
  - A 7 × 7 grid beat 5 × 5: cost 43.3 vs 53.8, speed 1.08 vs 0.99.
- **Design scoring.** Return cannot rank bodies here (it saturates), hence the speed score. Iterations shorter than one episode per lane bias scoring towards bodies that flip early.
- **Generalisation of the shared policy.** The boundary is mass, not distance in gene space: novel bodies inside the trained mass range worked (7/7, mean arrival 99%), outside it 3/7 (mean 63%).

C THROUGHPUT HISTORY: FROM THE ORIGINAL MJX-SAFETY-GYM TO NOW

Report only. Our code started from mjx-safety-gym 0.1.6 [43]. Steps of the chain were measured on different machines, environment counts and task versions, so the factors do not multiply exactly. Only the end-to-end comparison is a direct before/after measurement.

- **Engineering, same physics:** the JAX upgrade, environment counts, the Warp backend.
- **Changes to what is simulated:** condim 3 (Gym's setting), the minefield task (vases removed), walls, and the solver cap (an approximation, since replaced by Warp's converged solver).

D SIMULATOR EQUIVALENCE TESTS

- **One step from identical states.**
  - MuJoCo 3.3.2 vs 3.14 agree to float32 rounding when the solver converges.
  - With the 4/8 cap they differ by up to 1.4×10^{-3} in joint velocity.
  - Warp vs JAX agree to about 10^{-7}, including with per-lane bodies.
- **Vase cost via contact sensors** (MJX-JAX lacks subtree matching, so one sensor per vase–geom pair). It needs found *and* negative distance: 0 differing steps against the contact-list cost, while "found" alone over-counts by about 9% (5,735 vs 5,283).
- **Policy transfer.** A policy trained under the capped solver is 4.5% slower on the new stack (1.141 vs 1.195 m/s, 12 SE); cost, flips and exits are unchanged. Hence: never mix stacks in one comparison.
- **Learning check** (single nominal body, $d = 25$, 5 cm gate, mean of evaluations at 241–261M steps):

Stack	Return	Speed	Cost	Hz. steps	Flip %	OOB %
JAX capped (3090, 1.7 h)	10.89	1.45	27.4	70.8	2.6	1.9
Warp full (laptop, 46 min)	11.06	1.52	26.3	76.6	0.3	2.4

Table 7: Throughput history of the ant pipeline (report only).

When (2026)	Change	Measured effect
25 Jul	JAX 0.6.0 → 0.10.2	∼2.4× MJX stepping [CHECK]
9 Aug	first ant baseline (Goal task, vases, condim 6)	668–798 steps/s (3090, 256 envs)
13 Aug	contact dimension 6 → 3	1.82× (laptop, 128 envs)
15 Aug	Minefield task (no vases; hazards non-colliding)	3.13× (laptop, 256 envs)
17 Aug	1024 environments (3090 knee)	+31% over 512
22 Aug–26 Sep	corridor walls added, then removed	−27%, then +37%
27 Sep	solver capped at 4/8 (MJX-JAX)	3.5× (laptop)
4 Oct	MuJoCo Warp, full solver	3.1–3.3× vs capped JAX (3090)
4 Oct	4096 environments on Warp	1.75× vs 1024 (3090)
End to end	3090, ant	∼0.8k → 118–133k (≈150×); 220–233k at 4096 envs